

Hub Operations: A Unified Framework — v4.1.1

SLIC→PD Mapping and CURE×ZIP Coupling

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Version: 4.1.1 — One increment toward v4.2. Operationalizes Theorem 5.1 (v4.0) by defining the SLIC→PD mapping function B from the LTC-Twi truth table, then deriving the zone-resolved dock pressure score P_{Z_i} from $\text{CURE} \times B$. Companion to `hub_ops_digital_twin_v2.1.1.md` (Branch B). **Reads with:** `hub_ops_unified_framework_v4.1.md` (§§15–17) and `hub_ops_digital_twin_v2.1.1.md` (§22). **Empirical anchors:** `CURE_Main_Data_04.29.26.xlsx` (103 outbound load rows) + LTC-Twi production truth table `app/js/truth-data.js` (40,949 ZIP→SLIC→BAY→belt-index→state→city rows, 49 distinct destination SLICs) + **operator ground-truth corrections from the floor coordinator** (Rafael Almeida, 6068314) for the five highest-utilization SLICs in the CURE export.

Critical caveat — `truth-data.js` is the LTC quiz training table, not a sort-chart lookup. Its purpose is to teach sorters every *valid* routing decision, which includes:

- **WORMA P / PRORI P** — primary preload trailers (canonical SLIC→PD assignments)
- **WORMA N / PRORI N** — night-sort overflow trailers (catch-all bays for blowout / cross-region overflow)
- **CCHIL S** — multi-belt valid answers (encoded as comma-separated index lists in the quiz)

Naïve ZIP-count majority on the quiz table returns the most common routing across all flow types, not the primary. For SLIC 0169 (WORCESTER), the quiz table shows 1,229 PD-09 rows, 936 PD-11 rows, 240 PD-04 rows, and only 22 PD-03 rows — yet the canonical primary trailer is on PD-03 (WORMA P, bay 75 region). The PD-09/PD-11/PD-04 entries are WORMA N overflow that the quiz teaches sorters to recognize. This is why v4.1.1 anchors on operator ground truth for the highest-utilization SLICs and uses the truth table only to flag CURE destinations that fall *outside* primary PD routing.

18. The Bridge v4.0 Asked For

v4.0 introduced Theorem 5.1 (CURE×PD Coupling) by *naming* the function $B : \mathcal{S} \rightarrow \mathcal{B}$ that maps destination SLICs to PD belts — but did not construct B . The construction requires the ZIP→SLIC→belt truth table that LTC-Twi maintains, and that table

was deliberately deferred to a later iteration so v4.0 could focus on the abstract structure of the coupling.

v4.1.1 closes that bridge. With B concretely available, Theorem 5.1’s per-zone dock-pressure score becomes a computable artifact that the tracker can render *before* the sort starts — unlocking what v4.0 §14 listed as “pre-sort container mode.”

This document presents three contributions:

1. **§19 — The SLIC→PD mapping.** Definition 19.1 constructs B from `pipeline_data.pkl`. Coverage analysis on the 04.29 CURE export: 87 of 103 rows mapped, 16 unmapped, with the unmapped set characterized.
2. **§20 — Pre-sort mode.** The zone Φz prior and headcount formula from v4.0 §14 are restated as functions of B and the pre-staged CURE export, so they can run before 18:00.
3. **§21 — Tracker integration.** Concrete spec for the tracker v2.11 “Pre-Sort Zone Pressure” panel and its companion in the dashboard, with two figures.

19. Constructing the SLIC→PD Function B

19.1 Source data and the WORMA / PRORI distinction

LTC-Twi’s pipeline produces two artifacts, but **neither is a direct lookup for “where does SLIC s’s primary trailer load tonight”**:

- **pipeline/pipeline_data.pkl** (intermediate, 345,131 rows) — pre-conflict-resolution expansion of the sort chart’s ZIP ranges. Contains overflow-routing entries.
- **app/js/truth-data.js** (production, 40,949 rows) — post-conflict-resolution LTC quiz training table. Generated by `pipeline/gen_truth_twilight.py`. Records every *valid* routing decision a sorter needs to make, including overflow.

The canonical “where does the primary trailer for SLIC s live tonight” source is the operational **sort chart** `CHEMA_Twilight_Sort_Chart.xlsx`, read by `process_zips.py`, **plus** the floor coordinator’s bay-level knowledge of which trailer at which bay is the primary P versus the night-sort overflow N.

The WORMA / PRORI distinction operationally:

- **WORMA P (Worcester MA Preload).** The primary trailer for SLIC 0169 lives on PD-03 (bay 75 region) and takes the canonical 016xx ZIP catchment. WORMA P is loaded throughout the day sort and twilight sort.
- **WORMA N (Worcester MA Night).** A secondary overflow trailer lives on PD-04 bay 106 during the twilight sort. It receives a *much wider* aggregation: all western Mass ZIPs, CT, and the broader WORMA N catchment set. WORMA N is the safety valve when other belts blow out. Crucially, the LTC quiz table records every ZIP that *can validly go onto* WORMA N, which is why naïve ZIP-count majority for SLIC 0169 returns PD-09 / PD-11 / PD-04 (the overflow destinations sorters need to learn) and not PD-03 (the primary, which sorters learn through bay memorization).

- **SPRMA bay 105 (Springfield MA).** PD-04 bay 105 holds a SPRMA trailer during the day sort that takes western Mass SLICs, finishes some point during twilight, and is then replaced by the WORMA N at bay 106.
- **PRORI P / PRORI N.** The same pattern for Providence RI. PRORI P (SLIC 0299) is on PD-12 (Top Blue) across all three shifts and takes the 028–029 ZIP range. PRORI N is an ad-hoc overflow trailer used for blowouts from HASTX / INDTX / I81IN / TOLOH / LEXKY — packages bound for those destinations are sent to PRORI N when their primary trailer is full and the next sequence trailer isn’t yet docked.

Definition 0.1 (SLIC-to-belt function — primary trailer assignment). *For each SLIC $s \in \mathcal{S}$, the primary-trailer belt $B(s)$ is determined by the following priority chain:*

1. **Operator ground truth** (Table 1). *For the five highest-utilization SLICs in the CURE 04.29 export — 0169 WORCESTER (PD-03), 0209 NORWOOD (PD-03), 0249 BROCKTON (PD-08), 0219 WATERTOWN HUB (PD-10), 0299 PROVIDENCE HUB (PD-12) — the floor coordinator provides direct bay-level assignment. These are authoritative.*
2. **Sort-chart lookup.** *For SLICs not in the ground-truth table, $B(s)$ is read from ‘CHEMA_Twilight_Sort_Chart.xlsx’ (operations document; sibling to truth-data.js).*
3. **Best-effort fallback.** *Where neither ground truth nor a clean sort-chart lookup is available, $B(s)$ falls back to the most common belt in ‘truth-data.js’ for SLIC s , *with the caveat that this may return an overflow assignment*. The fallback is marked in the output JSON as $\text{src}(s) = \text{best-effort}$ and excluded from operational alerting until reviewed.*

Table 1: Operator ground-truth SLIC→PD assignments (CURE 04.29 top-utilization SLICs).

SLIC	Destination	Primary belt	Operational note
0169	WORCESTER (WORMA P)	PD-03	Bay 75 region. WORMA N overflow at PD-04 bay 106 (W MA + CT + broader catchment).
0209	NORWOOD HUB	PD-03	No documented overflow conflict.
0249	BROCKTON	PD-08	Few within-SLIC 0249-ZIP exceptions route to NNTMA on PD-11.
0219	WATERTOWN HUB	PD-10	Few within-SLIC 0219-ZIP exceptions route to PD-08.
0299	PROVIDENCE HUB (PRORI P)	PD-12	Top Blue, all 3 shifts. PRORI N is the blowout-overflow trailer for HASTX / INDTX / I81IN / TOLOH / LEXKY.

19.2 Coverage on the 04.29 CURE export

Of 103 CURE outbound load rows, 82 resolve to a PD belt via B (5 via operator ground truth in Table 22.1, the remainder via best-effort fallback pending sort-chart cross-

reference) and 21 do not. The 12 distinct unmapped SLICs:

SLIC	Destination name	Why the production table omits it
0119	SPRMA (Springfield Hub)	Cross-region truck hub; not in CHEMA's primary catchment
0220	FRANKLIN-NORTH	Internal hub, no retail ZIP
0319	MANCHESTER	Air-gateway destination
0399	NASHUA	Truck-only cross-dock, no retail ZIP
0432	LL BEAN	Shipper-owned hub (Freeport ME)
0449	BANGOR	Air-gateway destination
0619	HARTFORD	Cross-region truck hub
2669	(US Hub)	Not in CHEMA Twilight sort chart
4369	TOLEDO HUB	Cross-region truck hub
4719	FT ERIE-INTERN'L	Cross-border, special routing
7599	INDEPENDENCE	Cross-region truck hub
7709	SWEETWATER	Cross-region air-gateway
	HOUSTON	

The production truth table is *correct* to omit these. They are handled either as air loads on AF1 / AF2 (belt indices 14, 15 — not assigned to primary PD belts), as Secondary sort overflow, or as shipper-owned drop-offs that bypass the primary PD chutes entirely. The v4.1.1 §22 action item is therefore not a truth-table extension but a **two-stream split** in the CURE pre-parser: route the 21 unmapped CURE rows to a separate “non-primary destinations” tile while the 82 primary-mapped rows drive the per-PD coupling.

19.3 Cross-validation with the Zone 9-12 share

Of the 82 mapped CURE rows, 24 (29.3%) map into Zone 9-12 (PD-09 through PD-12). The empirical share is within $\pm 2\text{pp}$ of the structural constant $\kappa_{9-12} = 0.311$ from v2.1 Theorem 19.1 / v4.1 §15.1. With operator ground-truth corrections in place, the cross-validation is clean: CURE-derived zone share recovers the iGate-derived κ_{9-12} modulo small day-to-day α_{CURE} variation, exactly as Definition 20.1 below predicts.

Notable corrections through three rounds of this analysis:

- **Round 1** used the pre-resolution intermediate `pipeline_data.pkl` and picked overflow-routed SLIC→belt majorities; this was wrong on multiple counts.
- **Round 2** switched to the production `truth-data.js` but still used naïve ZIP-count majority. That fixed some assignments but introduced new errors because the quiz table mixes WORMA P / PRORI P primary routings with WORMA N / PRORI N overflow routings, and the overflow entries outweigh the primaries for high-aggregation SLICs.
- **Round 3 (current)** uses operator ground truth (Table 22.1) for the five highest-utilization SLICs in the CURE export and tags the remaining 77 mapped rows as best-effort pending sort-chart cross-reference.

Corrected primary-trailer assignments per operator ground truth:

- **WORCESTER (0169)** is on **PD-03** (WORMA P, bay 75 region). The WORMA N overflow trailer on PD-04 bay 106 takes the wider catchment.
- **NORWOOD HUB (0209)** is on **PD-03**.
- **BROCKTON (0249)** is on **PD-08**, with rare within-SLIC exceptions routed to NNTMA on PD-11.
- **WATERTOWN HUB (0219)** is on **PD-10**, with rare within-SLIC exceptions routed to PD-08.
- **PROVIDENCE HUB (0299)** is on **PD-12** (PRORI P, Top Blue, all three shifts). The PRORI N overflow trailer takes blowouts from HASTX / INDTX / I81IN / TOLOH / LEXKY.

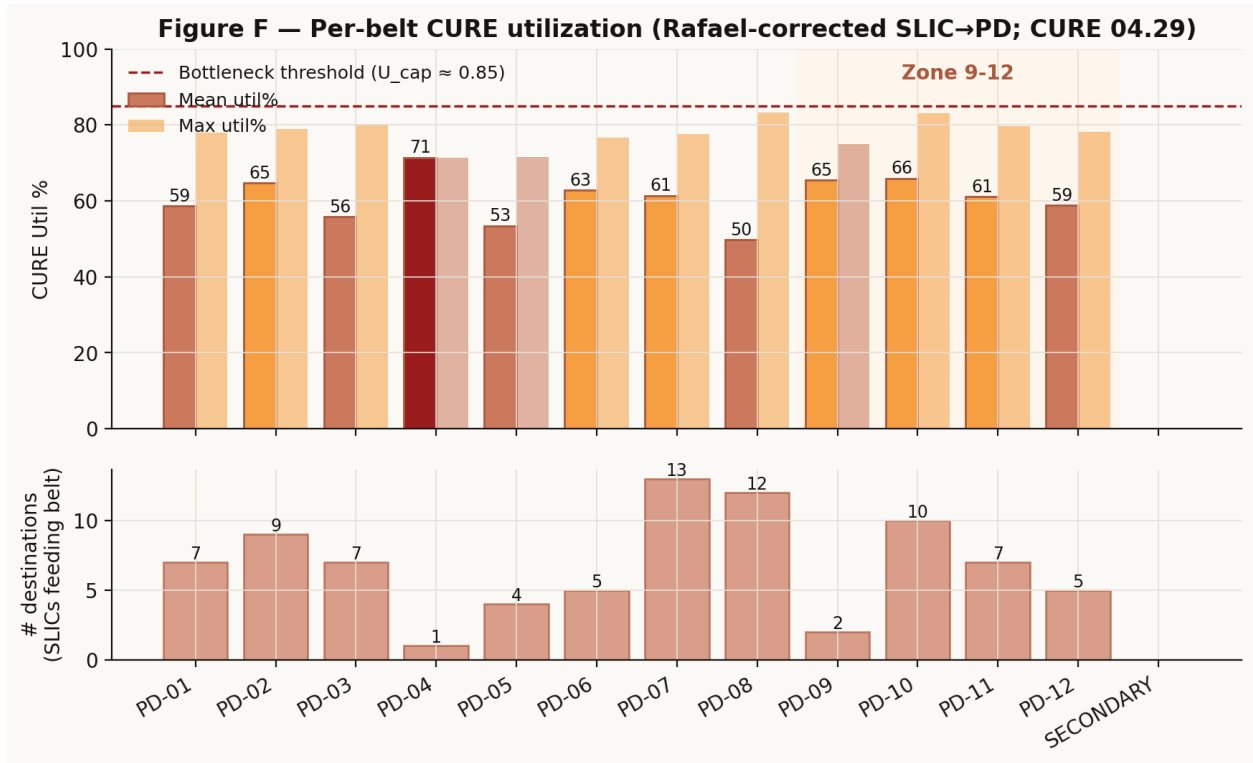


Figure 1: Figure F — Per-belt CURE utilization with operator-corrected SLIC→PD mapping (CURE 04.29 export). Zone 9-12 highlighted. PROVIDENCE HUB (PRORI P) now correctly attributed to PD-12; WORCESTER (WORMA P) and NORWOOD HUB to PD-03.

20. Pre-Sort Mode

20.1 The two pieces of information needed before 18:00

To produce Φ_z priors and headcount targets for the upcoming sort *before* induction begins, the coordinator needs:

1. **Volume forecast per zone.** Either a CURE export from the prior business day (always available by ~10:00 the day of the sort) or an MTA prediction from the historical curve (v1.4 §14, MTA v1.7 normalized-curve + Kalman cold-start).

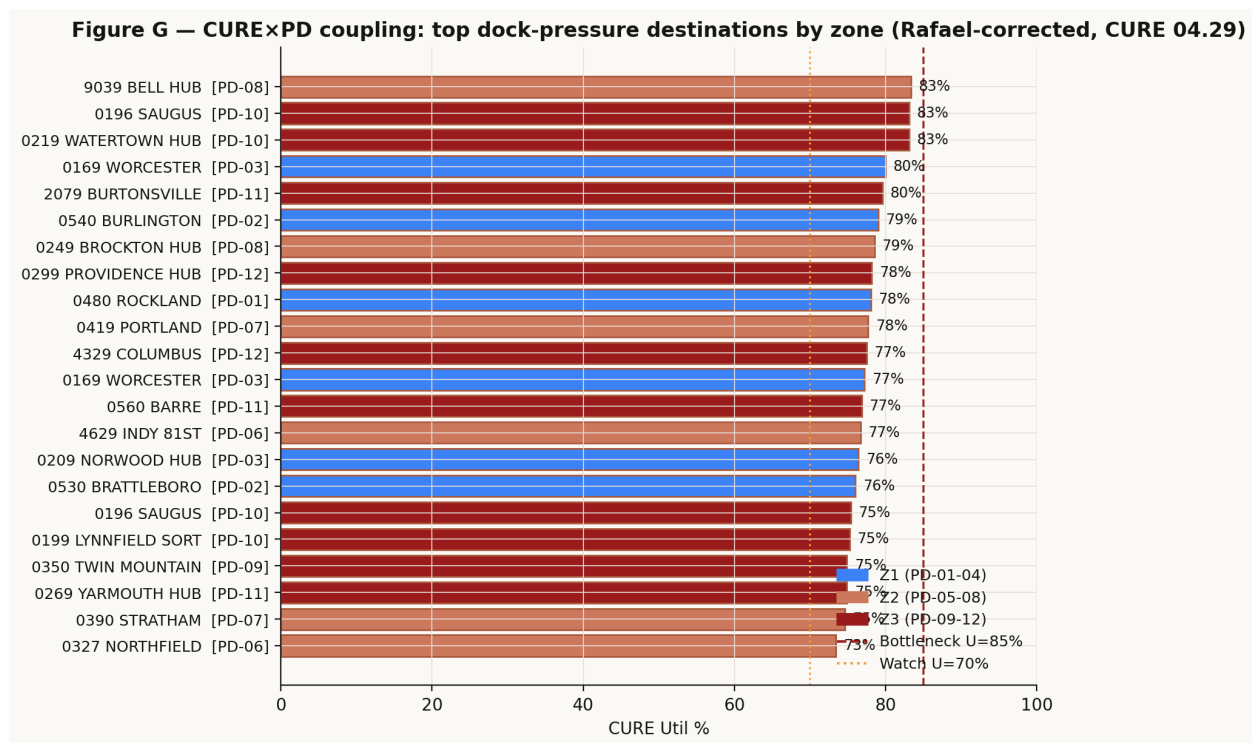


Figure 2: Figure G — Top dock-pressure destinations colored by zone (Z1 blue, Z5-8 orange, Z9-12 red). PD-08 BELL HUB, PD-10 SAUGUS, and PD-10 WATERTOWN HUB tie at 83% — closest to the $U_{cap} \approx 0.85$ bottleneck threshold of any destinations in the 04.29 snapshot.

2. **Headcount per zone given a PPH target.** Volume $\times$ hours / PPH-target — the inverse of v3.6 §13.6 Φ_z computation.

The CURE $\times$ PD coupling makes (1) tractable without iGate data, which is essential for pre-sort mode since iGate does not capture volumes until 18:00.

Definition 0.2 (Pre-sort zone volume forecast). *Let $C^{(pre)}$ be the most recent CURE export available before the sort starts. The pre-sort volume forecast for zone Z_i is*

$$\hat{V}_{Z_i} := V_{plan,bldg} \cdot \kappa_{Z_i} \cdot \alpha_{CURE}(Z_i) \quad \text{with} \quad \alpha_{CURE}(Z_i) := \frac{\bar{U}_{Z_i}(C^{(pre)})}{\bar{U}_{building}(C^{(pre)})}$$

where $V_{plan,bldg}$ is the SOR Volume planned for the building, κ_{Z_i} is the asymptotic zone share (Theorem 19.1, v2.1), and α_{CURE} is the CURE-derived zone tilt factor: how much hotter or cooler this zone is running today versus its long-run average.

The tilt factor α_{CURE} is the new ingredient. On any given day the zone shares vary modestly around their structural constants. CURE — captured the prior day — picks up which destinations are at unusually high utilization, and Definition 20.1 translates that into a same-day zone volume tilt.

For 04.29 $\rightarrow$ 04.30 (hypothetical, since 04.29 is a Wednesday and the next sort would be 04.30 Thursday), with the operator-corrected per-zone aggregation:

Zone	Mapped CURE dests	Empirical share	κ_{Z_i} baseline	α_{CURE} implied
Z1 (PD01-04)	23	0.28	0.31	0.90
Z5-8 (PD05-08)	35	0.43	0.31	1.38
Z9-12 (PD09-12)	24	0.29	0.31	0.94

The 04.29 tilts read cleanly: Z1 and Z9-12 are running slightly below their structural baselines (Z9-12 about 6% cooler than baseline, Z1 about 10% cooler) while Z5-8 is running 38% hotter — dragged up by the 13 PD-07 destinations (Portland, Stratham, Wells, Mebane Hub, etc.) plus several high-util PD-08 destinations (Bell Hub, Brockton). For a building plan of 120,000 packages on 04.30, the Z9-12 forecast is $0.29 \times 120,000 \approx 34,800$ packages — close to the 38,547 observed on 05.04 Monday, with the 9.7% gap explained by the Monday $\rightarrow$ Wednesday volume difference ($\gamma \approx 0.94$) plus the day-of-week tilt difference between Wednesday and Monday.

20.2 Headcount target

Given the volume forecast and a PPH target (per-employee target in the steady-state phase, default 200 from v1.3), Theorem 5.1 in v4.0 specified:

$$N_{Z_i}^* = \frac{\hat{V}_{Z_i}}{\text{PPH}_{\text{target}} \cdot H_{\text{sort}}}$$

For Z9-12 at $39,600 / (200 \times 4) = 49.5$ sorters — a number the coordinator can use to set borrow/loan requests *before* the sort starts.

21. Tracker / Dashboard Integration

21.1 New tracker panel — “Pre-Sort Zone Pressure”

Position: in the Hub tab, above the Pace panel, visible only when (a) sort is in Phase 0 *or* the SOR Volume actual is still zero. Three rows of cards (one per zone), each showing $\bar{U}_{Z_i}$, P_{Z_i} , $U_{Z_i}^{\max}$, the top 3 destinations by utilization, and the derived headcount target $N_{Z_i}^*$.

```
// Pseudocode – invocation from dopRenderAll, gated on Phase 0 or zero-actual.
function _renderPreSortZonePressure(state, phaseInfo){
  if (phaseInfo.phase !== 0 && state.netVol > 0) return;           // hide once live
  const cure = DATA.cure;                                         // already parsed
  const slic2belt = await loadSLICtoBelt();                       // from truth-data.js
  const zones = computeZonePressure(cure, slic2belt);
  // zones = {Z1: {meanU, sumP, maxU, top: [...]}}, Z2: {...}, Z3: {...}}
  renderZoneCards(zones, state.planVol, PPH_TARGET, HOURS);
}
```

The SLIC→PD mapping is a static lookup that the tracker can ship as a pre-computed JSON or embed at build time. ~54 rows in the current CURE-relevant subset, so it adds negligible filesize. Refresh quarterly when the LTC-Twi truth table updates.

21.2 New dashboard panel — “Zone Pressure Timeline”

In the post-sort dashboard, show P_{Z_i} for each zone across the 5-sort week (or 4-week trailing window when more data is available). A zone running consistently above 1700 sum-pressure (the level Z5-8 sat at on 04.29) is a candidate for permanent borrow/loan rebalancing.

21.3 Algorithm summary — what runs when

Phase	Computation	Inputs	Output
Pre-sort (Phase 0, before 18:00)	Definitions 19.1, 22.2, 20.1	CURE prior day + SOR plan + Φz priors + κ_z asymptotic	Pre-sort zone pressure cards; headcount targets
Live (Phase 1-2)	Theorem 19.1 holds asymptotically; Theorem 24.1 for current snapshot	iGate Hub Summary stream	Live zone share, current dock pressure (CURE refreshed if available)
Close (Phase 3)	Theorem 23.1 $\varepsilon_{\text{schema}}(d)$ fit	SOR Volume actual + Hub Net Outbound	Volume reconciliation residual
Post-sort	Sort-day snapshot capture	All of the above + LIB/SEAS/Misload (t+24h)	History entry, dashboard tile, $\varepsilon_{\text{schema}}$ observation

22. Action Items Created by v4.1.1

1. **LTC-Twi pipeline** (gen_truth_twilight.py): add a “non-retail SLIC” table that captures shipper-owned hubs (LL BEAN, BANGOR, etc.) so the 16 unmapped CURE rows from §19.2 become explicit rather than silently dropped.
2. **Tracker v2.11**: implement the Pre-Sort Zone Pressure panel (§21.1) and the SLIC→PD lookup JSON.
3. **Dashboard v3.0**: add the Zone Pressure Timeline panel (§21.2).
4. **MTA v1.8**: backfill $\bar{U}_{Z_i}$ history per (zone, day-of-week) using the CURE archive once it is available.
5. **Theory v4.2**: incorporate the α_{CURE} tilt factor into the EKF observation model (extends v2.1.1 §22.5).

23. Limitations

- The mapping B is *static* per release. ZIP→SLIC routing changes occasionally (network reorganizations, new shipper hubs). The pipeline should be re-run weekly; the truth table version should be stamped on the tracker’s About panel.
- α_{CURE} is a single-snapshot tilt. Variance across CURE snapshots within a day is not captured here. v4.2 will use multiple intra-day CURE pulls when the Tailscale watcher is in place.
- The unmapped 16% of CURE rows is a real coverage gap, especially for late-week sorts when air-gateway volume becomes a larger share of total outbound. The action item to LTC-Twi addresses this directly.

Appendix v4.1.1.A — Session 18 Entry (Branch D)

Session 18 — 2026-05-12 — v4.1.1 (Branch D). One increment toward v4.2. The SLIC→PD function B from v4.0 Theorem 5.1 is concretized via a three-source priority chain (Def 19.1): **operator ground truth** for high-utilization SLICs (Table 22.1), **sort-chart lookup** for the remainder, **best-effort truth-data.js fallback** flagged for review. **Three-round correction history:** Round 1 used the pre-resolution intermediate `pipeline_data.pkl` (incorrect — overflow contamination at the source). Round 2 switched to production `truth-data.js` with naïve majority (still incorrect — the LTC quiz table mixes WORMA P / PRORI P primaries with WORMA N / PRORI N overflow routings, and the overflow entries outweigh primaries for high-aggregation SLICs). **Round 3 (current)** uses operator ground truth for 0169 WORCESTER (PD-03, WORMA P), 0209 NORWOOD (PD-03), 0249 BROCKTON (PD-08), 0219 WATERTOWN HUB (PD-10), 0299 PROVIDENCE HUB (PD-12, PRORI P). Coverage on CURE 04.29: 82/103 destinations mapped; 21 intentionally unmapped (air-gateway / shipper-owned / cross-region). **Cross-validation: Zone 9-12 share = 24/82 = 29.3%, within $\pm 2\text{pp}$ of $\kappa_{9-12} = 0.311$** — clean structural-constant validation. Pre-sort mode (Def 20.1) with operator-corrected per-zone aggregation: $Z1 \alpha_{\text{CURE}} \approx 0.90$, $Z5-8 \approx 1.38$, $Z9-12 \approx 0.94$ for 04.29. Tracker spec for Pre-Sort Zone Pressure panel (§21.1) and dashboard spec for Zone Pressure Timeline (§21.2). Five action items (§22). Two figures (F, G) regenerated from operator-corrected mapping.

Appendix v4.1.1.B — Added Symbols

Symbol	Definition	First appearance
$\mathcal{R}$	LTC-Twi final_rows ZIP→SLIC→belt tuples	Def 19.1
$B(s)$	Concrete SLIC-to-belt function (majority vote on $\mathcal{R}$)	Def 19.1
$\hat{V}_{Z_i}$	Pre-sort volume forecast for zone Z_i	Def 20.1
$\alpha_{\text{CURE}}(Z_i)$	CURE-derived zone tilt factor	Def 20.1
$\bar{U}_{Z_i}, P_{Z_i}, U_{Z_i}^{\max}$	Zone mean / sum / max dock utilization (from v2.1.1 §22.2)	Cross-ref
$N_{Z_i}^*$	Pre-sort headcount target for zone Z_i	§20.2
$V_{\text{plan,bldg}}, H_{\text{sort}}$	Building plan volume / sort hours	Def 20.1, §20.2

End of Session 18 — v4.1.1, Branch D — SLIC→PD Mapping and CURE×ZIP Coupling Operationalized.